

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

## Chapter 7: Ratio and Proportion

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

### 1 Properties of proportion

1. Use the properties of proportion to solve  $\frac{6x^2 + 3x - 5}{3x - 5} = \frac{9x^2 + 2x + 5}{2x + 5}$ , where  $x \neq 0$ .  
ICSE 2025, 3 marks
2. Given  $\frac{(a+b)^3}{(a-b)^3} = \frac{64}{27}$ , find (a)  $\frac{a+b}{a-b}$  (b) the ratio  $a : b$ , using the properties of proportion.  
ICSE 2024, 4 marks
3. Use componendo and dividendo to solve  $\frac{\sqrt{2x+2} + \sqrt{2x-1}}{\sqrt{2x+2} - \sqrt{2x-1}} = 3$ . ICSE 2023, 3 marks
4. Find  $x : y$ , given  $\frac{x^2 + 2x}{2x + 4} = \frac{y^2 + 3y}{3y + 9}$ . ICSE 2020, 3 marks
5. If  $x = \frac{\sqrt{2a+1} + \sqrt{2a-1}}{\sqrt{2a+1} - \sqrt{2a-1}}$ , prove that  $x^2 - 4ax + 1 = 0$ . ICSE 2020, 4 marks
6. Use the properties of proportion to solve  $\frac{\sqrt{5x} + \sqrt{2x-6}}{\sqrt{5x} - \sqrt{2x-6}} = 4$ . ICSE 2019, 3 marks
7. Use the properties of proportion to solve  $\frac{2x + \sqrt{4x^2 - 1}}{2x - \sqrt{4x^2 - 1}} = 4$ , given that  $x$  is positive.  
ICSE 2018, 3 marks
8. Given  $\frac{7m + 2n}{7m - 2n} = \frac{5}{3}$ , use the properties of proportion to find (a)  $m : n$  (b)  $\frac{m^2 + n^2}{m^2 - n^2}$ .  
ICSE 2017, 3 marks

### 2 Proportion

9. A train moves at a uniform speed. It covers 60 m in 2 s, 90 m in  $x$  s and  $y$  m in 5 s. Find  $x$  and  $y$ . (a)  $x = 4, y = 150$  (b)  $x = 3, y = 100$  (c)  $x = 4, y = 100$  (d)  $x = 3, y = 150$  ICSE 2024, 1 mark
10. What number must be added to each of 4, 6, 8 and 11 so that the four numbers are in proportion?  
ICSE 2023, 3 marks
11. The numbers  $K + 3, K + 2, 3K - 7$  and  $2K - 3$  are in proportion. Find  $K$ . ICSE 2019, 3 marks

### 3 Continued proportion

12. Find the mean proportional between 4 and 9. (a) 4 (b) 6 (c) 9 (d) 36 ICSE 2023, 1 mark

13.  $b$  is the mean proportional between  $a$  and  $c$ . Show that  $\frac{a^4 + a^2b^2 + b^4}{b^4 + b^2c^2 + c^4} = \frac{a^2}{c^2}$ . ICSE 2017, 3 marks

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